

OPTICROP: SMART AGRICULTURAL PRODUCTION OPTIMIZATION ENGINE

IMPLEMENTATION DETAILS & SNIPPETS

1. Backend Routing Code (Flask)

The controller logic utilizes Flask routes to accept POST data, process it, and render the results template. Below is an excerpt of the prediction pipeline:

- Extract soil parameters from the HTML form request.
- Transform the input list using the StandardScaler pipeline.
- Perform inference using the Random Forest classifier.
- Predict estimated yield and categorise soil with the KMeans model.
- Append details to datasets/user_predictions.csv and render results.

2. Model Training Script (train_models.py)

The training pipeline loads datasets, splits them (80/20 train/test split), fits the StandardScaler, trains the classifiers, outputs charts, and serializes files using pickle.